EEX6335 – Compiler Design
EEX6363 – Compiler Construction

Day School – 3

by

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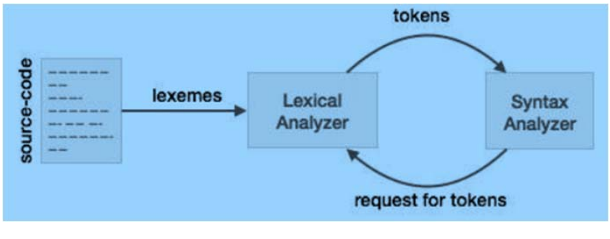
Let's consider detail analysis of syntax analyzer:



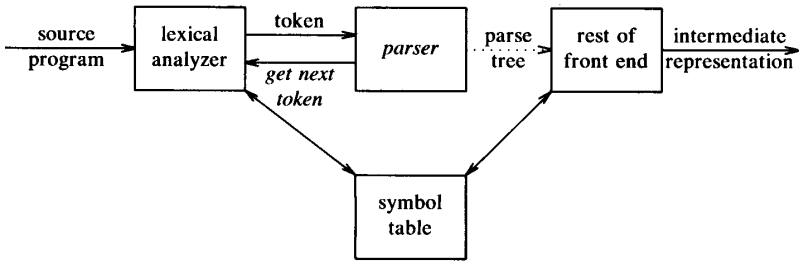
Syntax Analysis

- Context Free Grammars
- Parsing
  - Top-down -- *by left most derivation*
  - Bottom-up (shift-reduce) -- *by right most derivation in reverse*
- Attribute grammars

Syntax (Parser) Analysis

Parser:

If the string is generated from the given grammar, then the parser will generate a parse tree for that given string.



Input: sequence of tokens from lexer.

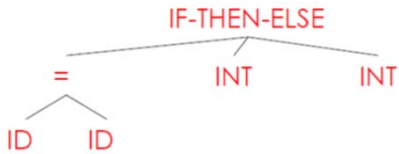
Output: parse tree of the program, but some parsers never produce a parse tree ...

E.g., Let us consider the if-then-else statement of a language:

if x = y then 1 else 2

Parser input → IF ID = ID THEN INT ELSE INT

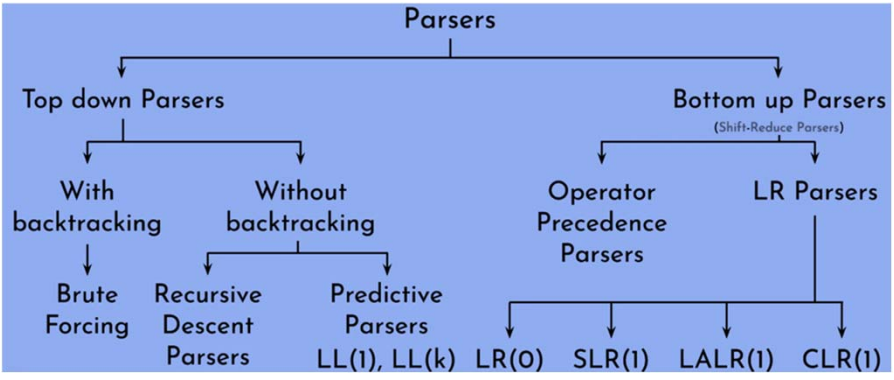
Parser output →



Comparison:

Phase	Input	Output
Lexer	String of characters	String of tokens
Parser	String of tokens	Parse tree

Types of Parsers



The Role of the Parser:

Not **all** strings of tokens are programs;

Thus, the parser must distinguish between valid and invalid strings of tokens. For that, we need:

- A language for describing valid strings of tokens;
- A method for distinguishing valid from invalid strings of tokens.

Programming language constructs have recursive structure:

E.g., An EXPR is

if EXPR then EXPR else EXPR

while EXPR loop EXPR …

Context Free Grammars

-- are a natural notation for this recursive structure.

Context Free Grammars

- A context free grammar consists of terminals, non-terminals, a start symbol, and productions.
- Terminals are the basic symbols from which strings are formed.
- Nonterminals are syntactic variables that denote sets of strings.
- One nonterminal is distinguished as the start symbol.
- The productions of a grammar specify the manner in which the terminal and non-terminals can be combined to form strings.
- A language that can be generated by a grammar is said to be a context-free language.
- A grammar lays down the rules to specify a language as opposed to a recognizer which identifies the strings belonging to a language.

RE relates to Context-Free Grammar

- Symbolically context-free grammar G can be represented as

CFG → G = (N, T (or $\sum$), P, S), where

N : Finite set of variables/non-terminals
T : Alphabet/Finite set of terminals
P : Finite set of rules/productions
S : Start symbol

$S \in N$
$N \cap T = \phi$
Rule : $A \rightarrow \omega$
$A \in N \quad \omega \in (N \cup T)^*$

with * represents a context-free language because we can write a CFG for it.

E.g., $G = (\{S\}, \{a\}, \{S \rightarrow \lambda, S \rightarrow aS\}, S)$

$A \rightarrow \lambda$; *A*-rule, *null*-rule, λ rule.

- *Context-freeness:*

An *A*-rule can be applied whenever *A* occurs in a string, irrespective of the context (that is, non-terminals and terminals around *A*).

- E.g., 1: The grammar with the following productions defines simple arithmetic expressions.

$$
\begin{aligned}
expr &\rightarrow expr\ op\ expr \\
expr &\rightarrow (\ expr\ ) \\
expr &\rightarrow -\ expr \\
expr &\rightarrow \mathbf{id} \\
op &\rightarrow + \\
op &\rightarrow - \\
op &\rightarrow * \\
op &\rightarrow / \\
op &\rightarrow \uparrow
\end{aligned}
$$

In this grammar, the terminal symbols are

id + − ∗ / ↑ ( )

The nonterminal symbols are *expr* and *op*, and *expr* is the start symbol. □

E.g. 2:

$$
\begin{aligned}
EXPR &\rightarrow \text{if EXPR then EXPR else EXPR} \\
&\mid \text{while EXPR loop EXPR} \\
&\mid \text{id}
\end{aligned}
$$

-Non-terminals are written in upper-case.

-Terminals are in lower-case.

-The start symbol is the left-hand side of the first production.

From CFG to Language

- One-step Derivation

$$\bullet uAv \Rightarrow uwv$$
$$A \rightarrow w$$

- w is derivable from v in CFG, if there is a finite sequence of rule applications such that:

$$v \Rightarrow w_1 \Rightarrow \ldots \Rightarrow w_n = w$$

E.g., From the given grammar, $E \rightarrow E{+}E \mid E * E \mid (E) \mid \text{id}$

derive the strings: id ∗ id and id ∗ id + id

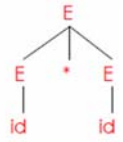
Solution:

We have, $E \rightarrow E+E \mid E*E \mid (E) \mid id$

first consider, id * id

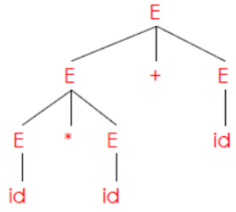
which can be derived as

$$
\begin{aligned}
& E \\
\rightarrow\ & E*E \\
\rightarrow\ & id*E \\
\rightarrow\ & id*id
\end{aligned}
$$



Next, id * id + id can be derived

$$
\begin{aligned}
& E \\
\rightarrow\ & E+E \\
\rightarrow\ & E*E+E \\
\rightarrow\ & id*E+E \\
\rightarrow\ & id*id+E \\
\rightarrow\ & id*id+id
\end{aligned}
$$



Notational Conventions

E.g., Let consider the following grammar:

```
E → EAE | (E) | -E | id
A → + | - | * | / | ↑
```

Derivations

- $E \Rightarrow -E$ is read "E derives -E"
- $E \Rightarrow -E \Rightarrow -(E) = -(\mathbf{id})$ is called a derivation of -(**id**) from E.
- If $A \rightarrow \gamma$ is a production and α and β are arbitrary strings of grammar symbols, we say $\alpha A\beta \Rightarrow \alpha\gamma\beta$.
- If $\alpha_1 \Rightarrow \alpha_2 \Rightarrow \ldots \Rightarrow \alpha_n$, we say α_1 derives α_n.

- $\Rightarrow$ means "derives in one step."
- $\underset{*}{\Rightarrow}$ means "derives in zero or more steps."
  - $\alpha \underset{*}{\Rightarrow} \alpha$
  - if $\alpha \underset{*}{\Rightarrow} \beta$ and $\beta \Rightarrow \gamma$ then $\alpha \underset{*}{\Rightarrow} \gamma$
- $\underset{+}{\Rightarrow}$ means "derives in one or more steps."
- If $S \underset{*}{\Rightarrow} \alpha$, where α may contain nonterminals, then we say that α is a sentential form.

Language, strings, …

- G: grammar, S: start symbol, L(G): the language generated by G.
- Strings in L(G) may contain only terminal symbols of G.
- A string of terminal *w* is said to be in L(G) if and only if $S \overset{+}{\Rightarrow} w$.
- The string *w* is called a sentence of G.
- A language that can be generated by a grammar is said to be a context-free language.
- If two grammars generate the same language, the grammars are said to be equivalent.

Example: $E \rightarrow EAE|(E)|-E|id$

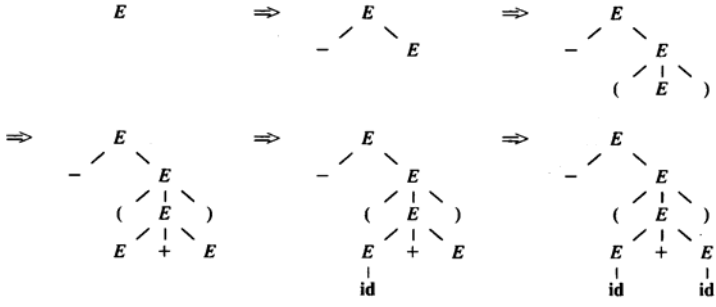
$A \rightarrow +|-|*|/|\uparrow$

The string -(id+id) is a sentence of the above grammar because

$E \Rightarrow -E \Rightarrow \ldots \Rightarrow -(E+E) \Rightarrow -(id+E) \Rightarrow -(id+id)$

Therefore, we can write $E \overset{*}{\Rightarrow} -(id+id)$

Now, lets draw the parse tree for this string



This is the parse tree for – (id + id)

Two Parse Trees

$E \rightarrow EAE|(E)|-E|id$

$A \rightarrow + | -| *| /| \uparrow$

Example 4.6. Let us again consider the arithmetic expression grammar The sentence **id+id*id** has the two distinct leftmost derivations:

$$\begin{aligned} E &\Rightarrow E+E \\ &\Rightarrow \mathbf{id}+E \\ &\Rightarrow \mathbf{id}+E*E \\ &\Rightarrow \mathbf{id}+\mathbf{id}*E \\ &\Rightarrow \mathbf{id}+\mathbf{id}*\mathbf{id} \end{aligned} \qquad \begin{aligned} E &\Rightarrow E*E \\ &\Rightarrow E+E*E \\ &\Rightarrow \mathbf{id}+E*E \\ &\Rightarrow \mathbf{id}+\mathbf{id}*E \\ &\Rightarrow \mathbf{id}+\mathbf{id}*\mathbf{id} \end{aligned}$$

with the two corresponding parse trees shown in Fig. 4.4.

-- *Dragon book*

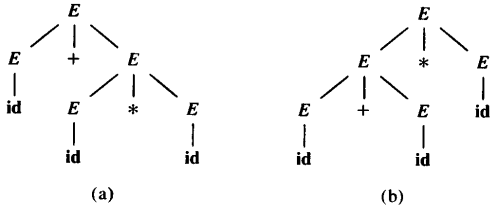


Fig. 4.4. Two parse trees for **id+id*id**.

Ambiguity

A grammar that produces more than one parse tree for some sentence is said to be *ambiguous*.

$$
\begin{aligned}
stmt &\rightarrow \textbf{if}\ expr\ \textbf{then}\ stmt \\
&\mid \textbf{if}\ expr\ \textbf{then}\ stmt\ \textbf{else}\ stmt \\
&\mid \textbf{other}
\end{aligned}
$$

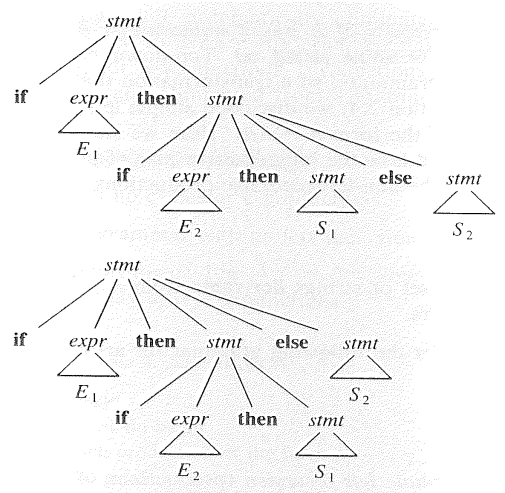


Fig. 4.6. Two parse trees for an ambiguous sentence.

Eliminating Ambiguity

Sometimes an ambiguous grammar can be rewritten to eliminate the ambiguity.

E.g., “match each else with the closest unmatched then”

$$
\begin{aligned}
stmt &\rightarrow matched_stmt \\
&\mid unmatched_stmt \\
matched_stmt &\rightarrow \textbf{if}\ expr\ \textbf{then}\ matched_stmt\ \textbf{else}\ matched_stmt \\
&\mid \textbf{other} \\
unmatched_stmt &\rightarrow \textbf{if}\ expr\ \textbf{then}\ stmt \\
&\mid \textbf{if}\ expr\ \textbf{then}\ matched_stmt\ \textbf{else}\ unmatched_stmt
\end{aligned}
$$

Eliminating Left Recursion

- A grammar is left recursive if it has a nonterminal A such that there is a derivation $A \overset{+}{\Rightarrow} A\alpha$ for some string α.
- $A \rightarrow A\alpha|\beta$ can be replaced by

$$A \rightarrow \beta A'$$

$$A' \rightarrow \alpha A'|\varepsilon$$

- $A \rightarrow A\alpha_1|A\alpha_2|\ldots|A\alpha_m|\beta_1|\beta_2|\ldots|\beta_n|$

$$A \rightarrow \beta_1 A'|\beta_2 A'|\ldots|\beta_n A'|$$

$$A' \rightarrow \alpha_1 A'|\alpha_2 A'|\ldots \alpha_m A'|\varepsilon$$

Algorithm: Eliminating Left Recursion

Input. Grammar G with no cycles or ϵ-productions.

Output. An equivalent grammar with no left recursion.

Method. Apply the algorithm in Fig. 4.7 to G. Note that the resulting non-left-recursive grammar may have ϵ-productions. □

1. Arrange the nonterminals in some order $A_1, A_2, \ldots, A_n$.
2. **for** $i := 1$ **to** n **do begin**
   for $j := 1$ **to** $i-1$ **do begin**
   replace each production of the form $A_i \rightarrow A_j\gamma$
   by the productions $A_i \rightarrow \delta_1\gamma \mid \delta_2\gamma \mid \cdots \mid \delta_k\gamma$,
   where $A_j \rightarrow \delta_1 \mid \delta_2 \mid \cdots \mid \delta_k$ are all the current A_j-productions;
   end
   eliminate the immediate left recursion among the A_i-productions
   end

Examples:

$$E \rightarrow E + T \mid T$$
$$T \rightarrow T * F \mid F$$
$$F \rightarrow (E) \mid \mathbf{id}$$

$$E \rightarrow TE'$$
$$E' \rightarrow +TE' \mid \epsilon$$
$$T \rightarrow FT'$$
$$T' \rightarrow *FT' \mid \epsilon$$
$$F \rightarrow (E) \mid \mathbf{id}$$

$S \rightarrow Aa|b$

$A \rightarrow Ac|Sd|\varepsilon$

S is left-recursive because

$S \rightarrow Aa \rightarrow Sda$

$A \rightarrow Ac|Aad|bd|\varepsilon$

$S \rightarrow Aa|b$

$A \rightarrow bdA'|A'$

$A' \rightarrow cA'|adA'|\ \varepsilon$

Simplifying Context Free Grammars

- Most of the time, some production rules in CFGs are not useful and are redundant.
- By simplifying CFGs we remove all these redundant productions from a grammar while keeping the transformed grammar equivalent to the original grammar.
- Two grammars are called equivalent if they produce the same language.
- Simplifying CFGs is necessary to later convert them into Normal forms such as CNF, GNF, etc.

Types of redundant productions: **useless, null** and **unit**

Useless productions –

The productions rules that can never take part in derivation of any string.

E.g., Consider the grammar:

S -> abS | abA | abB
A -> cd
B -> aB
C -> dc

S -> abS | abA
A -> cd

Null productions (λ) –

The productions of type 'A -> λ' are called λ productions (also called lambda and null productions).

These productions can only be removed from those grammars that do not generate λ (an empty string).

It is possible for a grammar to contain null productions and yet not produce an empty string.

E.g., Consider the grammar:

S -> ABCd
A -> BC
B -> bB | λ
C -> cC | λ

S -> ABCd | ABd | ACd | BCd | Ad | Bd |Cd | d
A -> BC | B | C
B -> bB | b
C -> cC | c

Unit productions –

The productions of type ‘A -> B’ are called unit productions.

E.g.,

S -> Aa | B
A -> b | B
B -> A | a

S -> Aa
A -> b
B -> a

S -> Aa
A -> b | a
B -> a

S -> Aa
A -> b | a
B -> a | b

S -> Aa | b | a
A -> b | a
B -> a | b

S -> Aa|b|a
A -> b|a

Do you have any questions/clarifications?

… few minutes.

Top-Down Parsing:

- Recursive-descent parsing
- Predictive parsers
- Non-recursive predictive parsing
- FIRST and FOLLOW
- Construction of predictive parsing table
- LL(1) grammars
- Error recovery in predictive parsing

Recursive-Descent Parsing

- Top-down parsing can be viewed as an attempt to find a leftmost derivation for an input string.
- It can also view as an attempt to construct a parse tree for the input string from the root and creating the nodes of the parse tree in preorder.

Grammar:

$S \rightarrow cAd$
$A \rightarrow ab \mid a$

Input string
w = *cad*

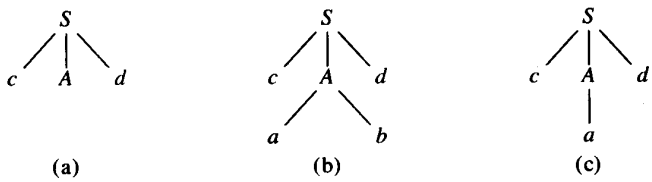


Fig. 4.9. Steps in top-down parse.

Implement of Recursive-Descent Parsing

After removing left-recursion,

$S \rightarrow cAd$
$A \rightarrow ab \mid a$

$S \rightarrow cAd$
$A \rightarrow aA'$
$A' \rightarrow b \mid \varepsilon$

```
S()
{
        if (look_ahead == 'c')
        {
                match('c');
                A();
                match('d');
        }
}
```

```
A()
{
        if (look_ahead == 'a')
        {
                match('a');
                A'();
        }
}
```

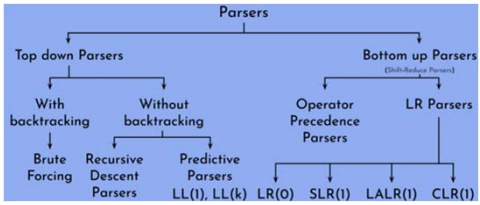
```
A'()
{
        if (look_ahead == 'b')
        {
                match('b');
        }
        else
                return;
}
```

```
match(char i)
{
        if (look_ahead == i)
                look_ahead = getchar();
        else
                printf("Error ");
}
```

```
main()
{
        S();
        if (look_ahead == ' $')
                printf("Parsing Successfully");
}
```

Input string
w = *cad* $

Predictive Parsers



- By carefully writing a grammar, eliminating left recursion, and left factoring the resulting grammar, we can obtain a grammar that can be parsed by a recursive-descent parser that needs <u>no backtracking</u>, i.e., a predictive parser.

$stmt \rightarrow$ **if** $expr$ **then** $stmt$ **else** $stmt$
$\mid$ **while** $expr$ **do** $stmt$
$\mid$ **begin** $stmt_list$ **end**

$S \rightarrow cAd$
$A \rightarrow ab \mid a$

$S \rightarrow cAd$
$A \rightarrow aA'$
$A' \rightarrow b \mid \varepsilon$

Predictive Parsers

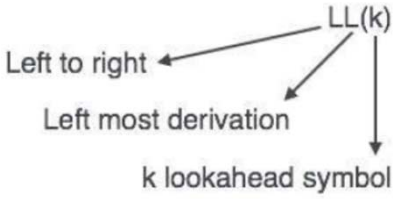
- Recursive-descent parsing is a top-down method of syntax analysis in which we execute a set of recursive procedures to process the input.
- A procedures is associated with each nonterminal of a grammar.
- Predictive parsing is what in which the look-ahead symbol unambiguously determines the procedure selected for each nonterminal.
- The sequence of procedures called in-processing the input implicitly defines a parse tree for the input.

LL(1) Grammars

- A grammar whose parsing table has no multiply-defined entries is said to be LL(1).
- LL Parser accepts LL grammar. LL grammar is a subset of context-free grammar but with some restrictions to get the simplified version, in order to achieve easy implementation.
- LL grammar can be implemented by means of both algorithms namely, recursive-descent or table-driven.

Stands for LL(1):

- First L: scanning from left to right
- Second L: producing a leftmost derivation
- 1: using one input symbol of lookahead at each step to make parsing action decision.



FIRST and FOLLOW sets

- If α is <u>any string</u> of grammar symbols, FIRST(α) is the set of terminals that begin the strings derived from α. If $\alpha \overset{*}{\Rightarrow} \varepsilon$ then ε is also in FIRST(α).
- FOLLOW(A), for non-terminal A, is the set of terminals a that can appear immediately to the right of A in some sentential form, i.e., the set of terminals a such that there exists a derivation of the form $S \Rightarrow \alpha A a \beta$ for some α and β.
- If A can be the rightmost symbol in some sentential form, the $ is in FOLLOW(A).

$S \rightarrow aABC$
$A \rightarrow b$
$B \rightarrow c$
$C \rightarrow d$

FIRST(S): S → a A B C

FIRST(A): A → b

FIRST(B): B → c

FIRST(C): C → d

$S \rightarrow ABC$
$A \rightarrow a$
$B \rightarrow b \mid \varepsilon$
$C \rightarrow c$

S $ → A B C $; A → ε, C → c

FOLLOW(S): { $ }
FOLLOW(A): { b, c }
FOLLOW(B): { c }
FOLLOW(C): { $ }

Properties of LL(1)

- No ambiguous or left recursive grammar can be LL(1).
- Grammar G is LL(1) **iff** whenever $A \rightarrow \alpha|\beta$ are two distinct productions of G and:
  - For no terminal, *a* do both α and β derive strings beginning with *a*.

$$\text{FIRST}(\alpha) \cap \text{FIRST}(\beta) = \varnothing$$

  - At most one of α and β can derive the empty string.
  - If $\beta \Rightarrow \varepsilon$, the α does not derive any string beginning with a terminal in FOLLOW(A).

$$\text{FIRST}(\alpha\text{FOLLOW}(A)) \cap \text{FIRST}(\beta\text{FOLLOW}(A)) = \varnothing$$

Predictive Parsing Program

Input. A string w and a parsing table M for grammar G.

Output. If w is in $L(G)$, a leftmost derivation of w; otherwise, an error indication.

Method. Initially, the parser is in a configuration in which it has $\$S$ on the stack with S, the start symbol of G on top, and $w\$$ in the input buffer. The program that utilizes the predictive parsing table M to produce a parse for the input is shown in Fig. 4.14. □


```
set ip to point to the first symbol of w$;
repeat
    let X be the top stack symbol and a the symbol pointed to by ip;
    if X is a terminal or $ then
        if X = a then
            pop X from the stack and advance ip
        else error()
    else    /* X is a nonterminal */
        if M[X, a] = X → Y1Y2 · · · Yk then begin
            pop X from the stack;
            push Yk, Yk-1, . . . , Y1 onto the stack, with Y1 on top;
            output the production X → Y1Y2 · · · Yk
        end
        else error()
until X = $    /* stack is empty */
```

Parsing Table, M

Grammar:

$$
\begin{aligned}
E &\rightarrow TE' \\
E' &\rightarrow +TE' \mid \epsilon \\
T &\rightarrow FT' \\
T' &\rightarrow *FT' \mid \epsilon \\
F &\rightarrow (E) \mid \mathbf{id}
\end{aligned}
$$

Input: `id + id * id`

NONTERMINAL	INPUT SYMBOL					
	id	+	*	(	)	$
E	$E \rightarrow TE'$			$E \rightarrow TE'$		
E'		$E' \rightarrow +TE'$			$E' \rightarrow \epsilon$	$E' \rightarrow \epsilon$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow \epsilon$	$T' \rightarrow *FT'$		$T' \rightarrow \epsilon$	$T' \rightarrow \epsilon$
F	$F \rightarrow \mathbf{id}$			$F \rightarrow (E)$		

Moves Made by Predictive Parser

STACK	INPUT	OUTPUT
$\$E$	id + id * id$	
$\$E'T$	id + id * id$	$E \rightarrow TE'$
$\$E'T'F$	id + id * id$	$T \rightarrow FT'$
$\$E'T'\mathbf{id}$	id + id * id$	$F \rightarrow \mathbf{id}$
$\$E'T'$	+ id * id$	
$\$E'$	+ id * id$	$T' \rightarrow \epsilon$
$\$E'T+$	+ id * id$	$E' \rightarrow +TE'$
$\$E'T$	id * id$	
$\$E'T'F$	id * id$	$T \rightarrow FT'$
$\$E'T'\mathbf{id}$	id * id$	$F \rightarrow \mathbf{id}$
$\$E'T'$	* id$	
$\$E'T'F*$	* id$	$T' \rightarrow *FT'$
$\$E'T'F$	id$	
$\$E'T'\mathbf{id}$	id$	$F \rightarrow \mathbf{id}$
$\$E'T'$	$	
$\$E'$	$	$T' \rightarrow \epsilon$
$	$	$E' \rightarrow \epsilon$

Do you have any questions/clarifications?

… few minutes.

Bottom-Up Parsing and LR Parsers

- Shift-reduce parsing attempts to construct a parse tree for an input string beginning at the leaves (bottom) and working up towards the root (top).
- “Reducing” a string w to the start symbol of a grammar.
- At each reduction step a particular substring machining the right side of a production is replaced by the symbol on the left of that production, and if the substring is chosen correctly at each step, a rightmost derivation is traced out in reverse.

Example

- Grammar:

```
S → aABe
A → Abc|b
B → d
```

- Reduction:

abbcde
aAbcde
aAde
aABe
S

$$S \underset{rm}{\Longrightarrow} aABe \underset{rm}{\Longrightarrow} aAde \underset{rm}{\Longrightarrow} aAbcde \underset{rm}{\Longrightarrow} abbcde$$

LR(k) Parsers

- The LR parser is a non-recursive, shift-reduce, bottom-up parser.
- It uses a wide class of context-free grammar which makes it the most efficient syntax analysis technique.
- L: left-to-right scanning of the input
- R: constructing a rightmost derivation in reverse
- k: the number of input symbols of lookahead that are used in making parsing decisions.

- There are three widely used algorithms available for constructing an LR parser:

SLR(1) – Simple LR Parser:
- Works on smallest class of grammar;
- Few number of states, hence very small table;
- Simple and fast construction;

LR(1) – LR Parser:
- Works on complete set of LR(1) Grammar;
- Generates large table and large number of states;
- Slow construction;

LALR(1) – Look-Ahead LR Parser:
- Works on intermediate size of grammar;
- Number of states are same as in SLR(1);

LR Parsing

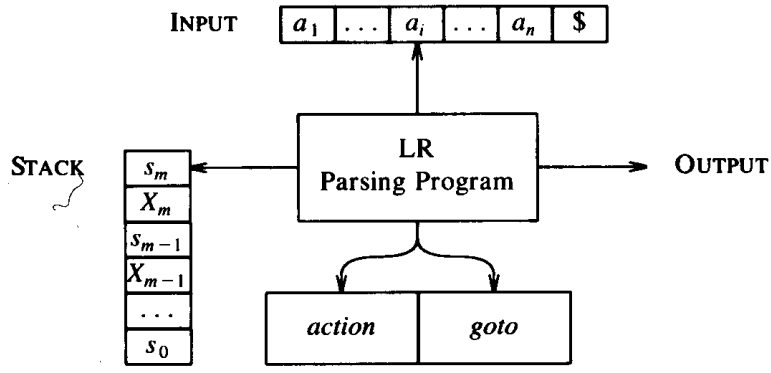
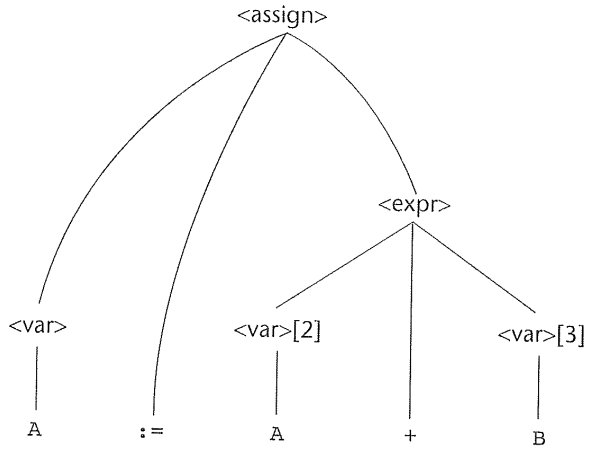


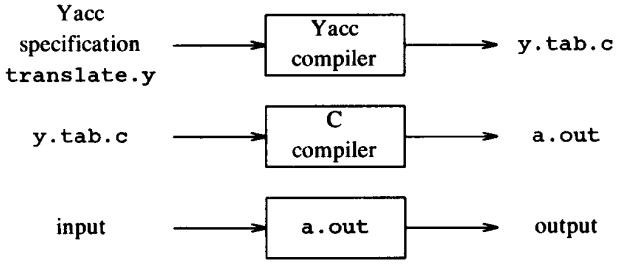
Fig. 4.29. Model of an LR parser.

E.g., A parser tree for A := A + B



- Tools for Syntax Analysis (YACC)

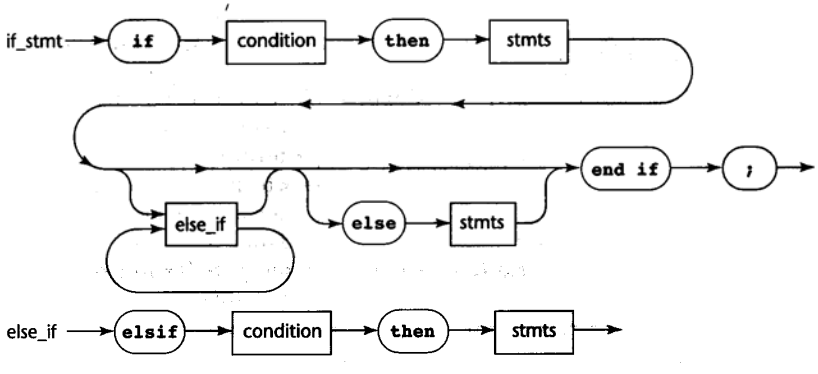
```
declarations
%%
translation rules
%%
supporting C-routines
```



Syntax Graphs

- A graph is a collection of nodes, some of which are connected by lines (edges).
- A directed graph is one in which the lines are directional.
- A parse tree is a restricted form of directed graph.
- Syntax graph is a directed graph representing the information in BNF rules.

Syntax Graph of “if “ statement



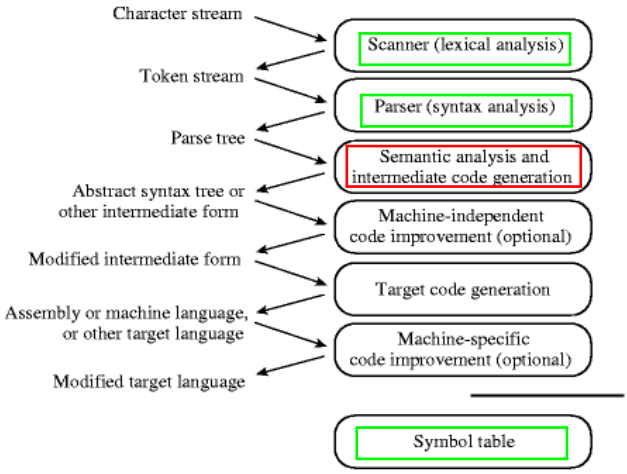
<if_stmt> ⟶ **if** <condition> **then** <stmts> {<else_if>}

[**else** <stmts>] **end if**

<else_if> ⟶ **elsif** <condition> **then** <stmts>

Phases of Compilation

Up to last DS, . . .



What's next?

We would like to:

- perform additional checks to increase guarantees of correctness;
- transform the program from the source language into the target one, and according to precisely defined semantic rules;

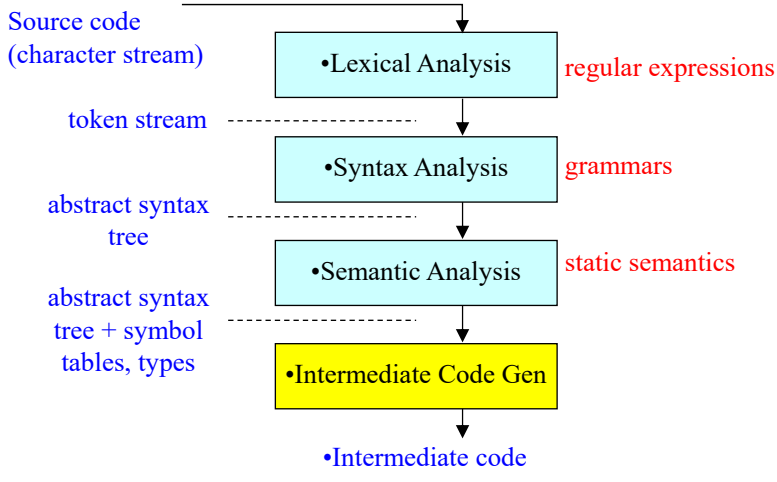
Additional Checks:

There are many additional checks that can be performed to increase correctness of code:

- Coherent (clear idea) usage of variables
  - definition-usage;
  - type;
- Existence of unreachable code blocks, . . .

Here, it is mainly focused on the mechanisms for type checking and generation of intermediate code

Where We Are...



What's next?

- After semantic analysis, we have a number of options to choose from:
  - generate object code directly from the parse.
  - generate intermediate code, and then generate object code from it.
  - generate an intermediate abstract representation, and then generate code directly from it.
  - generate an intermediate abstract representation, generate intermediate code, and then the object code.
- All these options have one thing in common:

  •-- they are all based on syntactic information gathered/aggregated/ processed/verified in the syntactic-semantic analysis.

•**Intermediate representations**

- Synthetize the syntactic information gathered during the parse, generally in the form of a tree or directed graph.
- Intermediate representations enable high-level code optimization.
- Three ways of intermediate representation:
  - Syntax tree
  - Postfix notation
  - Three address code
- In our project, we use an Abstract Syntax Tree (AST) as an intermediate representation.

•**Intermediate code**

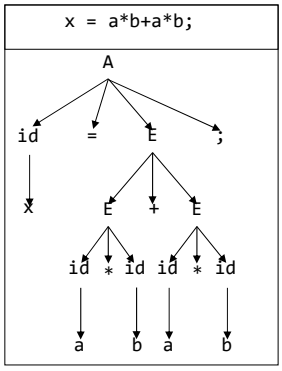
- is a low-level coded (text) representation of the program, directly translatable to object code.
- Intermediate code enables low-level, architecture-dependent optimizations.
- We don't have intermediate code in our project.

For quick reminding …

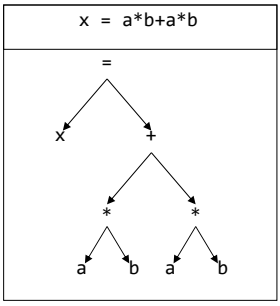
- **Parse trees** (concrete syntax trees) emphasize the grammatical structure of the program, based on the exact concrete syntax of the grammar.
- **Abstract syntax trees** emphasize the actual computations to be performed. They do not refer to the actual non-terminals defined in the grammar, nor to tokens that play no role in defining the translated program, hence their name.
- When implementing an LL parser, there is generally substantial differences between the parse tree and the corresponding abstract syntax tree.

- A parse tree contains <u>grammar-specific nodes</u>, e.g. non-terminal symbols, punctuation, etc. These are not necessary for further processing. Abstract syntax trees <u>do not include them</u>, hence the name *abstract*.

Parse Tree

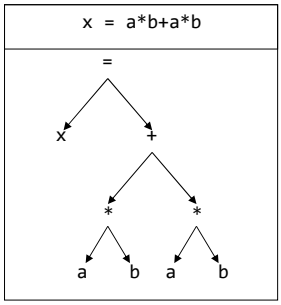


Abstract Syntax Tree

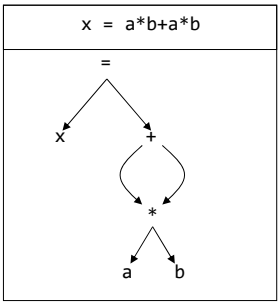


- **Directed acyclic graphs** (DAG) are a relative of syntax trees: they are used to show the syntactic structure of valid programs in the form of a graph with the following restrictions:
  - Edges between nodes can only be **unidirectional**.
  - There cannot be **cycles** in a DAG, i.e. no path can ever lead twice to the same node.
  - Presence of a **root node**.
- For example, in a DAG representation, the nodes for repeated variables and expressions are merged into a single node.
- DAGs are more complex to build and use than syntax trees, but easily allows the implementation of a variety of **optimization** techniques by avoiding redundant operations.

- Abstract Syntax Tree

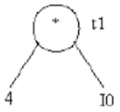


- Directed Acyclic Graph

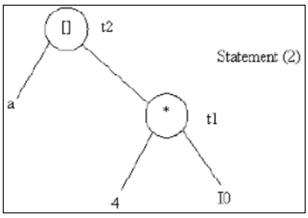


Example: Consider the block of three-address statements:

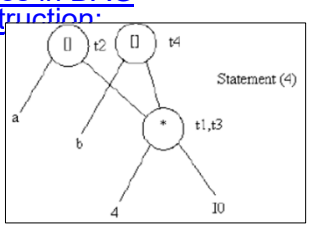
1. t1 := 4* i
2. t2 := a[t1]
3. t3 := 4* i
4. t4 := b[t3]
5. t5 := t2*t4
6. t6 := prod+t5
7. prod := t6
8. t7 := i+1
9. i := t7
10. if i<=20 goto (1)

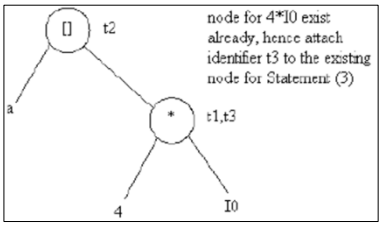


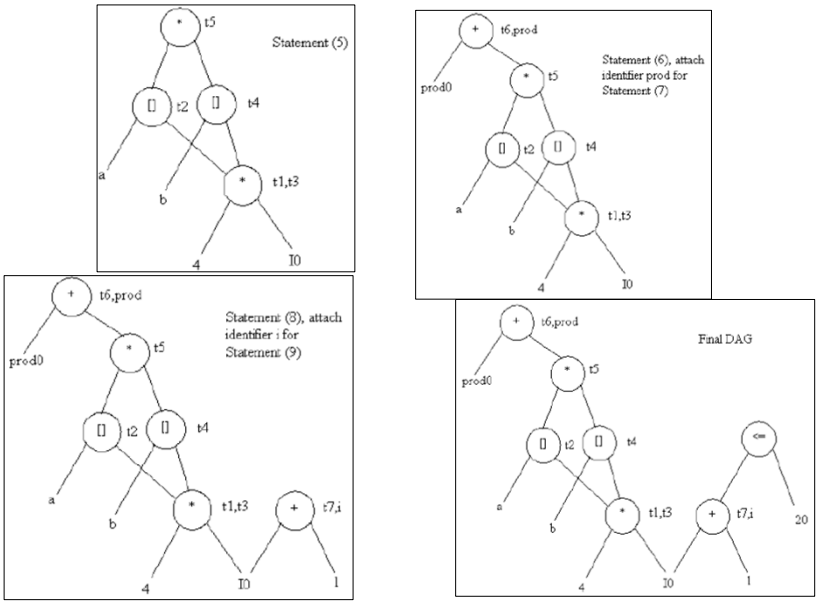
Statement (1)



- Stages in DAG Construction:







Thank You

END.